

EasySave v3.0

Technical Documentation

1. Overview

EasySave v3.0 adds parallel job execution, real-time job control (pause/resume/stop), priority file management, large file bandwidth limiting, CryptoSoft single-instance enforcement, Docker-based log centralization, and full GUI language switching. All changes are backward-compatible with v1.0, v1.1, and v2.0.

2. Architecture

2.1 Projects

Project	Type	Purpose
EasySave	Console app (.exe)	Interactive menu, CLI mode, job management (v1.1)
EasySave.GUI	WPF app (.exe)	Graphical interface using MVVM (v3.0)
EasyLog	Class library (.dll)	Daily log (JSON/XML), state file, Docker forwarding
CryptoSoft	Console app (.exe)	XOR encryption, single-instance via mutex

2.2 New and Modified Files — EasySave

File	Change
Models/AppSettings.cs	Added PriorityExtensions, MaxLargeFileKb, DockerLogMode, DockerLogUrl
BackupManager.cs	Parallel execution via Task/Thread, pause/stop/resume per job
Services/BackupStrategyBase.cs	Priority check, large file semaphore, pause/stop signals
Services/BusinessSoftwareDetector.cs	Now triggers pause instead of stop

2.3 New and Modified Files — EasySave.GUI

File	Change
ViewModels/MainViewModel.cs	Parallel run, pause/resume/stop commands, progress binding, Docker config
ViewModels/SettingsViewModel.cs	Added PriorityExtensions, MaxLargeFileKb, DockerLogMode, DockerLogUrl
Views/SettingsWindow.xaml	New sections: Priority Extensions, Large File Limit, Docker Centralization
MainWindow.xaml / .cs	Added x:Name attributes, ApplyLanguage() called on settings save

2.4 New and Modified Files — EasyLog

File	Change
DailyLogger.cs	Added Docker HTTP forwarding via HttpClient. SetDockerConfig() method added.

DockerLogMode.cs	New enum: Local, Remote, Both
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2.5 New Files — docker-server/

File	Purpose
server.py	Python HTTP server listening on port 5050. Appends log entries to daily JSON file.
Dockerfile	Builds a python:3.11-slim image running server.py
docker-compose.yml	Exposes port 5050, mounts ./logs volume for persistent log storage

3. Features

3.1 Parallel Backup

Each job runs in its own thread. A shared semaphore limits simultaneous large-file transfers to one at a time. Priority extensions are enforced globally: no non-priority file transfers while any job has priority files pending. Pause uses a ManualResetEventSlim; stop uses a CancellationToken.

3.2 Docker Log Centralization

DailyLogger.SendToDocker() posts each LogEntry as JSON to the configured URL via HttpClient. The Docker server (server.py) appends entries to /logs/YYYY-MM-DD.json inside the container. The ./logs volume maps to docker-server/logs on the host for persistence.

3.3 Priority File Management

Before transferring each file, BackupStrategyBase checks if any job globally has priority files remaining. If so, non-priority files wait. Priority extensions are read from AppSettings.PriorityExtensions.

3.4 Large File Bandwidth Limit

A static SemaphoreSlim(1,1) is shared across all jobs. Before transferring a file larger than AppSettings.MaxLargeFileKb, a job acquires the semaphore. It is released after the transfer. Smaller files bypass the semaphore entirely.

4. Settings Fields (v3.0 additions)

Field	Type	Default	Description
PriorityExtensions	List<string>	[]	Extensions copied before all others
MaxLargeFileKb	int	1024	Files larger than this (KB) cannot transfer in parallel
DockerLogMode	enum	Local	Local, Remote, or Both
DockerLogUrl	string	http://localhost:5050/log	URL of the Docker log server

5. Configuration & File Locations

File	Path
Job configuration	%APPDATA%\EasySave\config.json
App settings	%APPDATA%\EasySave\settings.json

Daily log (local)	%APPDATA%\EasySave\Logs\YYYY-MM-DD.{json xml}
State file	%APPDATA%\EasySave\State\state.json
Docker logs	docker-server/logs/YYYY-MM-DD.json (on host)

6. Minimum Requirements

Component	Requirement
Operating system	Windows 10 / Windows Server 2016 or later
Runtime	.NET 8.0 Runtime
Docker (optional)	Docker Desktop with WSL2 backend
Disk space	10 MB for application + sufficient space on target drive
Permissions	Read on source directory, write on target directory